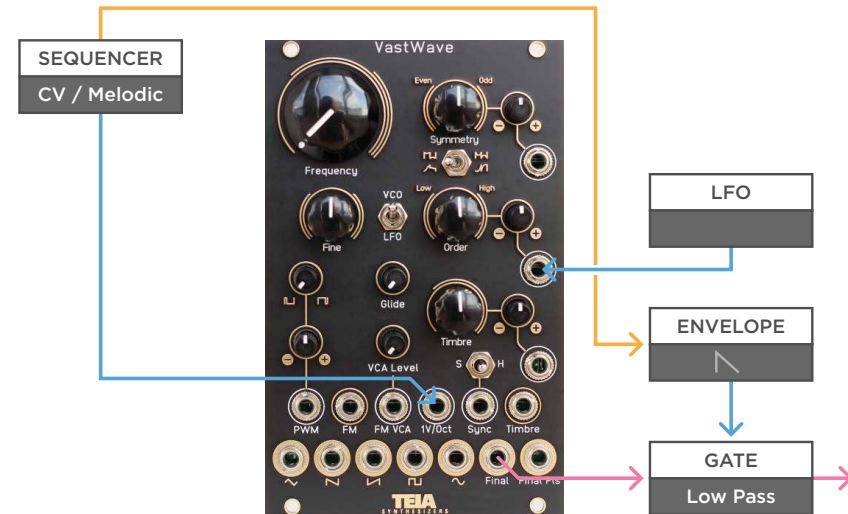


Oscillator + Wavefolder = VastWave

Teia Synthesizers

— Audio
— Control Voltage
— Trigger / Gate

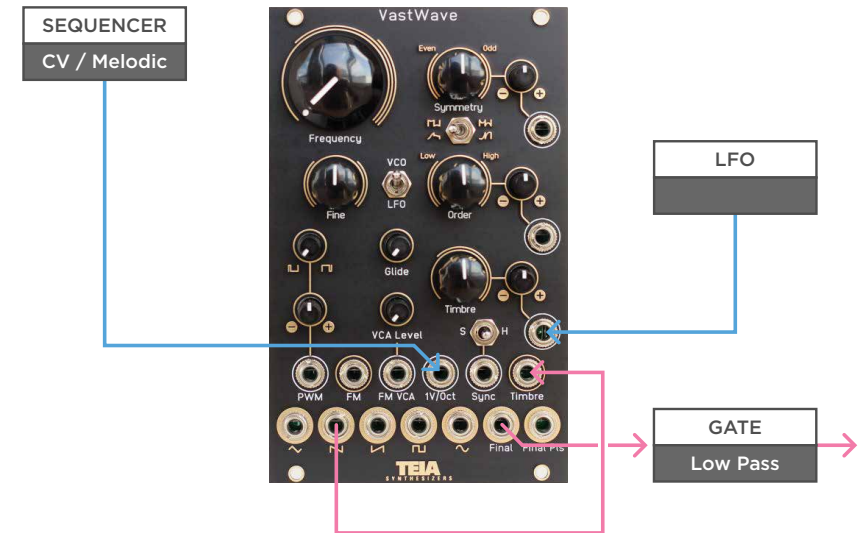


(a)

a - Let's set up a simple voice by taking the Final output into a low pass gate, use a sequencer to tune the oscillator, and trigger an envelope opening the gate. With just a single LFO slightly modulating the Order, you create a simple voice with some light movement, that can be taken from gentle to more extreme by hand.

MONOTRAIL

TECH TALK



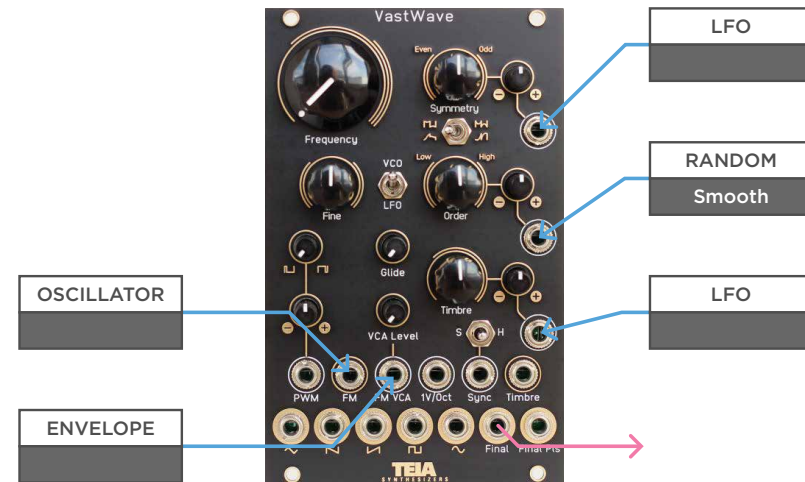
(b)

b - Because the timbre section has an external input, you can manually patch other waveforms into the folder. I really like the harmonics the folder adds to a saw wave for example. With some slow modulation to the Timbre it can create lovely organ style sounds.

Oscillator + Wavefolder = VastWave

Teia Synthesizers

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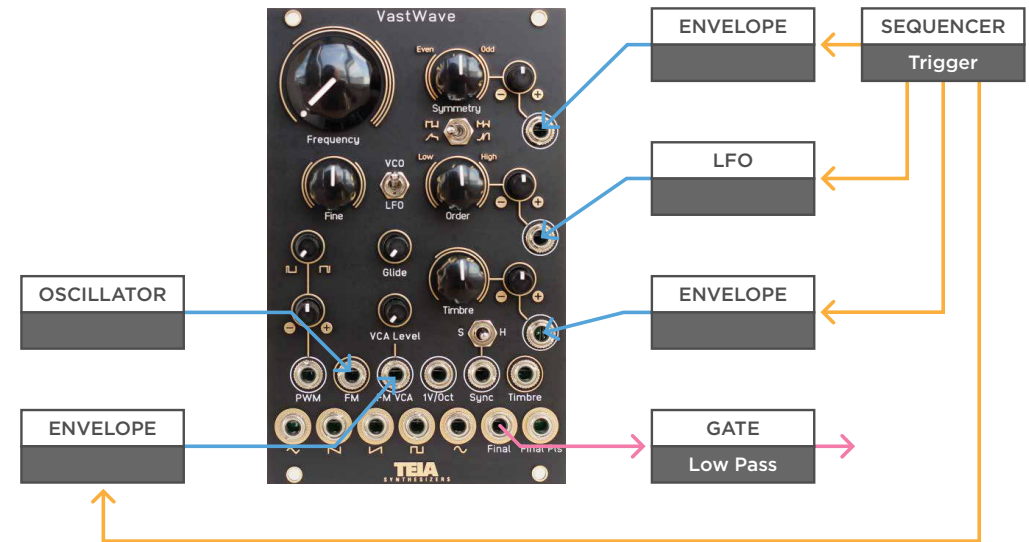


(a)

a - In this patch the oscillator is used to create a constant steady drone. To add some subtle texture a somewhat faster LFO to the symmetry works well. Then a slow LFO is modulating the folder, changing the timbre section from gentle to more aggressive waveforms. And finally, a slow random voltage is added to crossfade between the high and low sections, adding organic unpredictable movement to the patch. You can add to this by modulating the FM input. You can take a steady oscillator and send a sinewave to the FM input, then you can use something like a looping envelope to modulate the internal VCA, creating bursts of FM.

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(b)

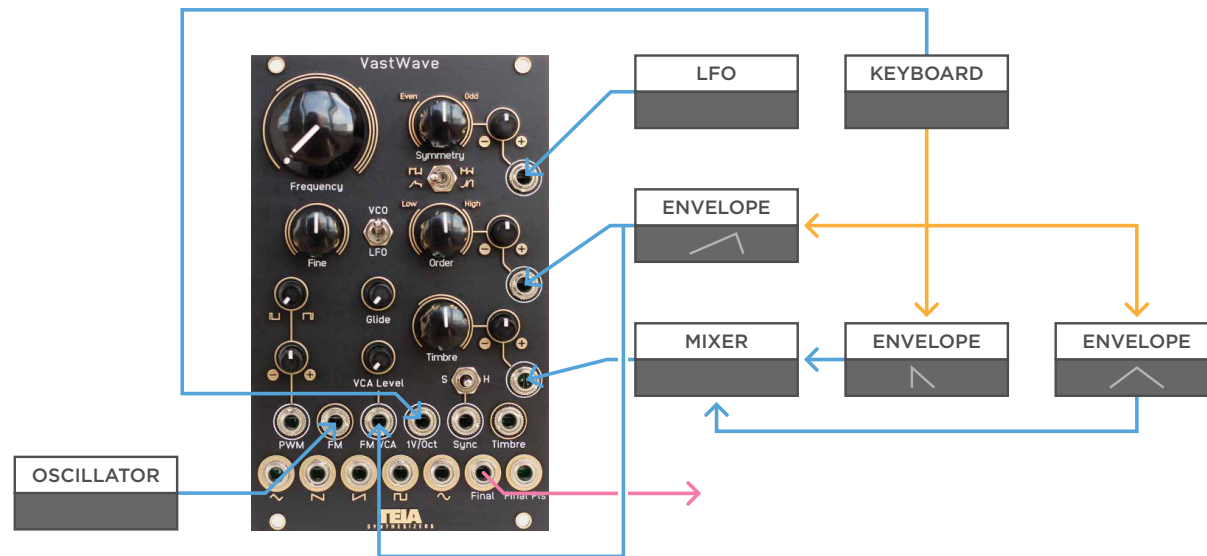
b - Instead of slow and random changes you can use a clock and clock divider, or pattern sequencer, to trigger envelopes or retrigger LFOs, and create tempo synced rhythmic patterns. Experiment by sending different signals in different amounts to different destinations. Again a second oscillator is used for FM.

Oscillator + Wavefolder = VastWave

Teia Synthesizers

MONOTRAIL TECH TALK

- Audio
- Control Voltage
- Trigger / Gate



(a)

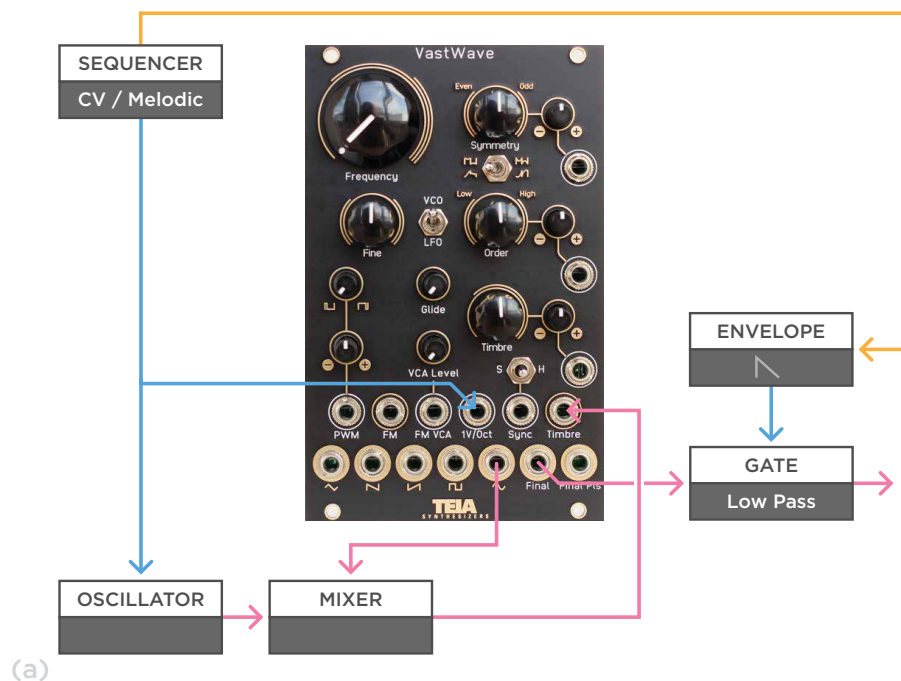
a - If you're more into playing tunes by hand you can use a similar setup. Use a keyboard or controller to tune the oscillator and use the gate output to gate or trigger some envelopes. For example, use a mixer to send a fast short envelope to the fold input, to add a transient to your sound.

You can add a second more subtle triangle shape envelope to create a sort of delay like sound. Then use a slow rising envelope to the Order, and the FM amount, and perhaps a slow LFO to the symmetry, and you have a very playable voice.

Oscillator + Wavefolder = VastWave

Teia Synthesizers

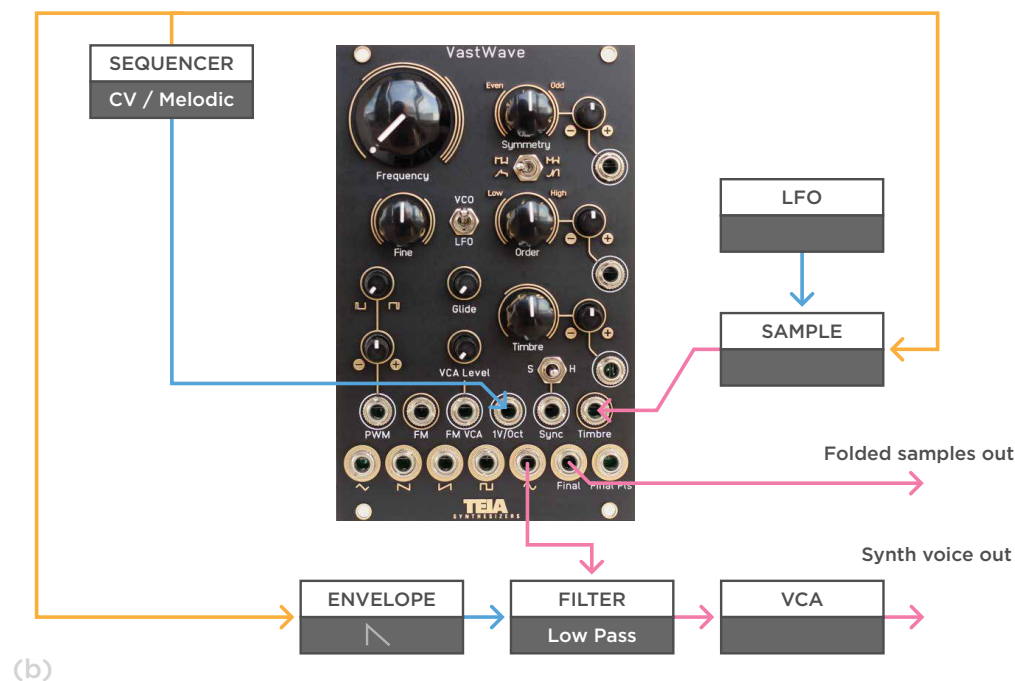
— Audio
— Control Voltage
— Trigger / Gate



a - The external input for the timbre section opens up a lot of possibilities. For example, you can add a second oscillator to add complexity to the folded sound. Use a mixer to mix the second oscillator with an output of the VastWave, and send the mix to the timbre input. When making drones you can use a voltage controlled mixer or VCA to control the amount of second oscillator to the mix. But this trick also works great in a simple melodic voice. In that case, make sure to send a 1v/oct signal to both oscillators.

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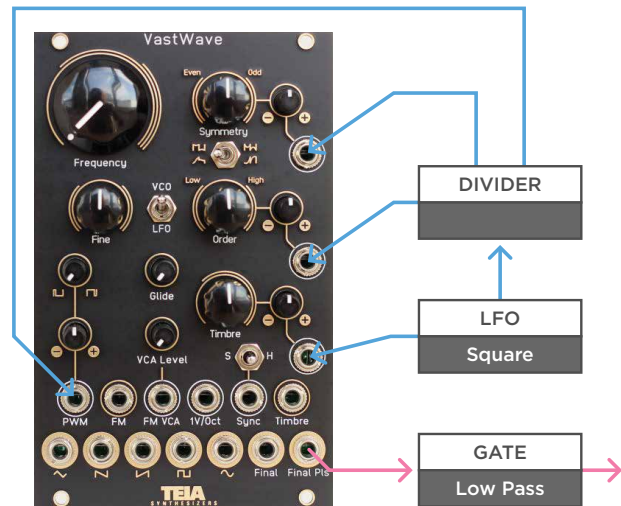
b - Wave folders can have a nice effect on a lot of sounds, for example a drum bus. Let's use a sample player triggered by a melodic sequencer, with an LFO to select different samples. If you set the Order knob completely to the Timbre section the folder can be used as a separate function, in this case, folding just the drums. So in this setup you can still use the oscillators basic outputs to create a regular independent synth voice, using the same sequencer.

Oscillator + Wavefolder = VastWave

Teia Synthesizers

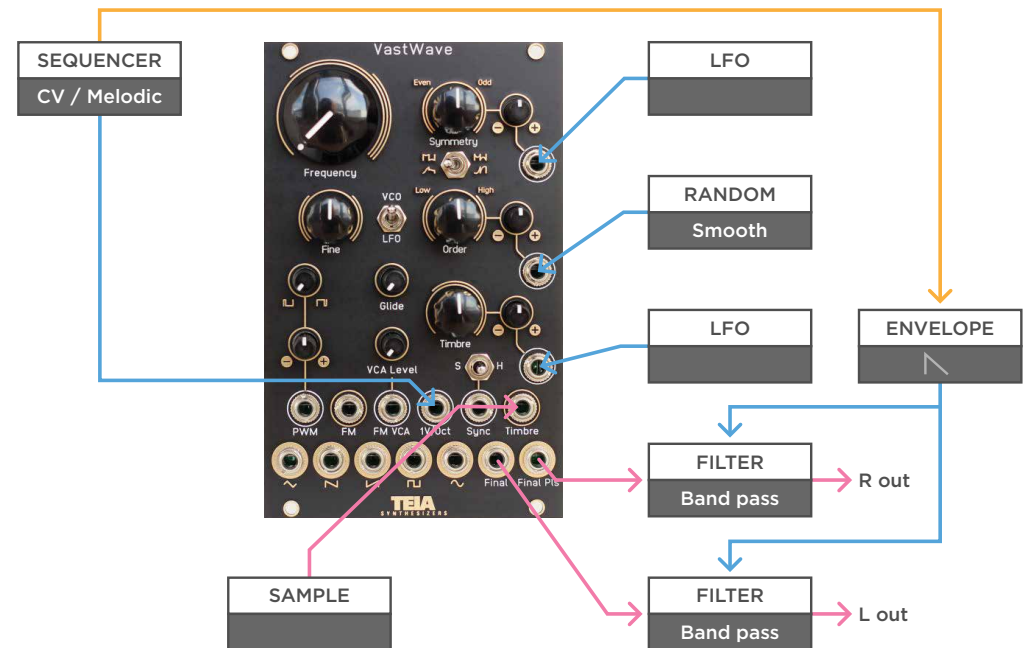
MONOTRAIL TECH TALK

- Audio
- Control Voltage
- Trigger / Gate



(a)

a - The pulse out also adds extra possibilities. In this patch a square wave LFO is used to modulate the Timbre. The LFO also drives a clock divider, and different patterns are used to modulate the Order, Symmetry, and pulse with modulation. The pulse output is passed through a low pass gate.



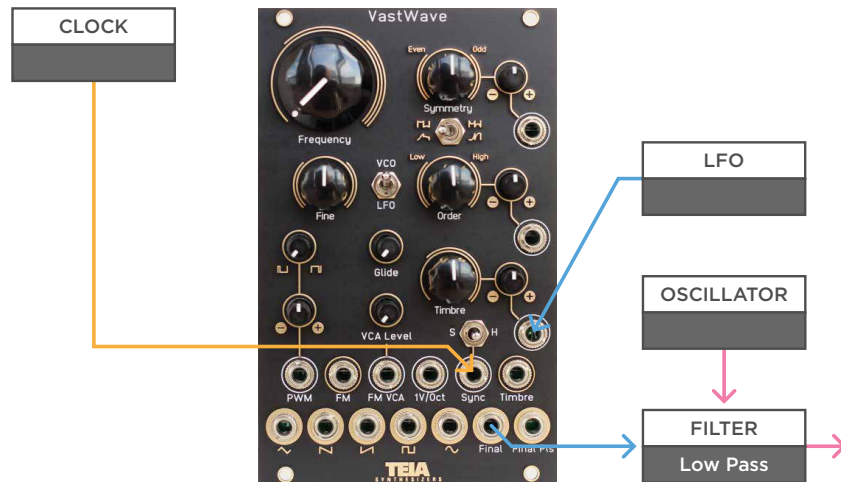
(b)

b - The pulse output is a lot harsher than the sine out, but by sending each to an individual filter and filtering the pulse out a bit more, you can balance that out. The folder parameters influence both sounds simultaneously so you can easily create stereo drones and voices by adding some modulation, and using a sequencer to trigger an envelope, and modulate both filters simultaneously. You can experiment with feeding samples to the timbre input to create sort of bit crushed noises and sounds. You can use the Order control to crossfade between the sample, and the internal oscillator.

Oscillator + Wavefolder = VastWave

Teia Synthesizers

- Audio
- Control Voltage
- Trigger / Gate

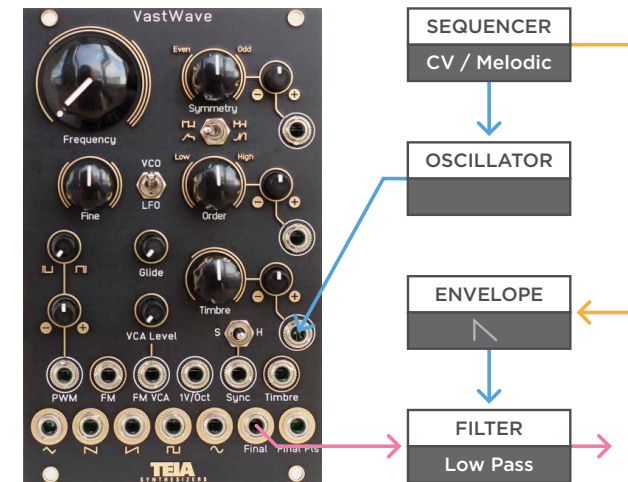


(a)

a - By using the wave folder on basic analog shapes in LFO mode, they basically becoming looping patterns, that can be used to modulate anything. For example a filter in a simple voice. You can use a clock on the hard sync input to tempo sync the pattern. By slowly modulating the folder, or LFO frequency, you create wave morphing variations to the looping pattern.

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(b)

b - Another trick is to use low frequency waveforms as the input of a folder, but add another oscillator to fold the LFO into audible tuned shapes. Just add a sequencer for example, to tune the modulation oscillator, and experiment by having that oscillator modulate different folding parameters at audio rates. In this setup it's actually the folded LFO you're listening to, and the signal responsible for the slow changes in sound over time.